

ASSIGNMENT 1

TIME SERIES MODELLING CASE STUDY

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GitHub Link: <https://github.com/Sindhu809/Time-Series-Analysis--24071473.git>

SYNOPSIS

This study is not a competition of the most advanced algorithm, but a practical question: which forecast is reliable when the origin is fixed and the next 104 weeks are truly unknown? German actual load from Open Power System Data was cleaned on hourly resolution and lowered to entire Sunday-ending weekly means. The design then set four benchmarks: SARIMA, weather-and-holiday SARIMAX, a recursively operated Random Forest and an hourly LSTM on the same test horizon. The annual-repeat rule was the winner regarding the RMSE, MAE and MAPE. The closest was the Random Forest (RMSE 3,153.71 MW) but still 5.54% worse. The statistical models suffered from a greatly differenced specification and a shifting amount of demand, while the LSTM lost long-horizon stability as its synthetic hourly outputs were frequently fed back. Weather and holidays improved SARIMAX relative to SARIMA by 21.20% but did not overcome seasonal persistence and used realized temperature, thus its outcome is conditional, not totally operational.

1 DATA USAGE

Germany's ENTSO-E actual-load field spans January 2015 to September 2020. Invalid timestamps and numeric values were coerced, duplicates were removed, a complete UTC index was constructed and negative demand was screened. Hourly MW values were averaged—not summed—so daily and weekly quantities retain the interpretation of mean demand. Weeks below 98% of 168 expected hours were excluded, leaving 299 observations.

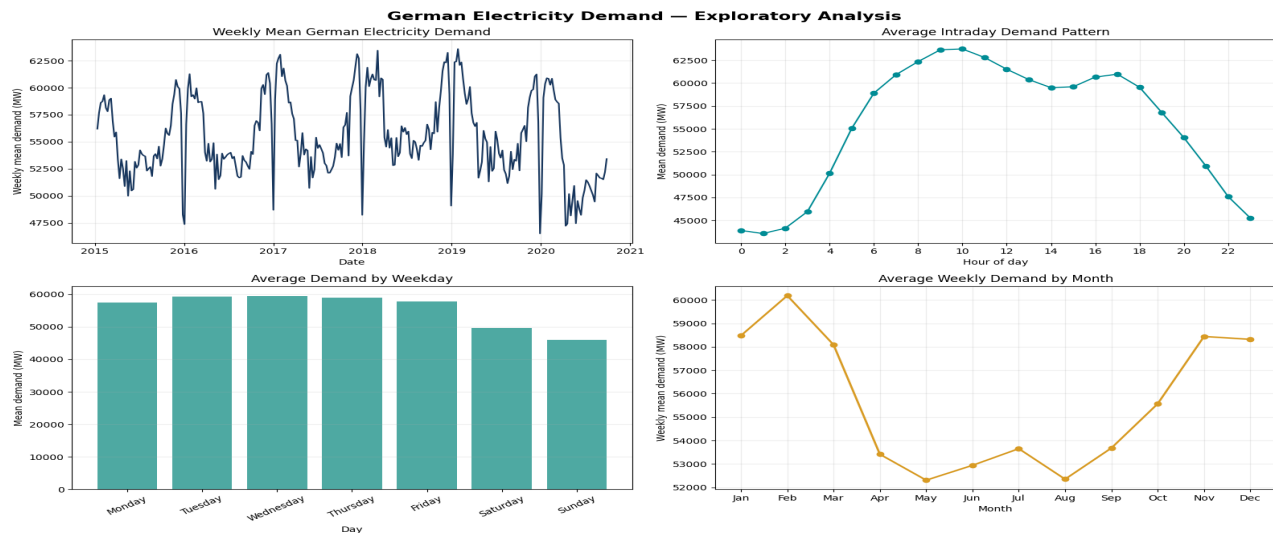


Figure 1. Demand structure across time, hour, weekday and month. The annual winter–summer contrast is visible before any model is fitted.

The modelling hypotheses are defined by the exploratory panels. Strong winter peak and summer trough stimulate 52-week benchmark and seasonal ARIMA terms. The intraday shape and the weekday–weekend difference suggest testing an hourly time series model. The late weakness in the record cautions that a model trained on previous levels may overpredict the test period. So, the same figure both gives a motivation for complexity and explains why complexity can fail. Why weekly averages? The comparison is usually weekly. The horizon of choice is two years. Aggregation suppresses the hourly noise while keeping the annual cycle. Aggregation also avoids the LSTM to receive an easier evaluation since it is trained at a finer frequency.

2 DIAGNOSIS & STATIONARITY

The null hypotheses of ADF and KPSS test are in opposing direction, so they were utilized as complimentary tests. The original weekly series passed both tests (ADF $p=0.0020$; KPSS $p=0.1000$). First differencing and the combination of first and seasonal differencing likewise passed, whereas seasonal differencing alone failed both tests. This rules out a mechanical claim that differencing is automatically required.

Transformation	ADF statistic	ADF P Value	KPSS statistic	KPSS P Value	Result
Original	-3.9005	0.0020	0.1632	0.1000	Stationary
First difference	-7.0351	<0.0001	0.0313	0.1000	Stationary
Seasonal difference	-1.8042	0.3784	1.4238	0.0100	Non-stationary
First and seasonal	-7.6900	<0.0001	0.0346	0.1000	Stationary

The substantive reason for $s=52$ is the ACF signal at year. But it does not justify all combinations of d and D . The assignment-constrained search considered 147 regular orders, $p,d,q \in \{0,6\} \times \{0,1,2\} \times \{0,\dots,6\}$, with seasonal order fixed at $(1,1,1)$ [52]. AIC chooses $(2,2,6)$ with SARIMA $(2,2,6) \times (1,1,1)$ [52]. The original series was already stationary; the fitted covariance matrix was near singular and test performance eventually deteriorated. This is reported faithfully but treated critically. The evidence therefore shows over-differencing and unstable uncertainty rather than a reliable gain from the lowest AIC. Diagnostic discipline Stationarity tests define stochastic behavior; they do not rank predictions. AIC rates the fitted candidates on training data; it does not guarantee a decreased blind-test error. The results support both claims.

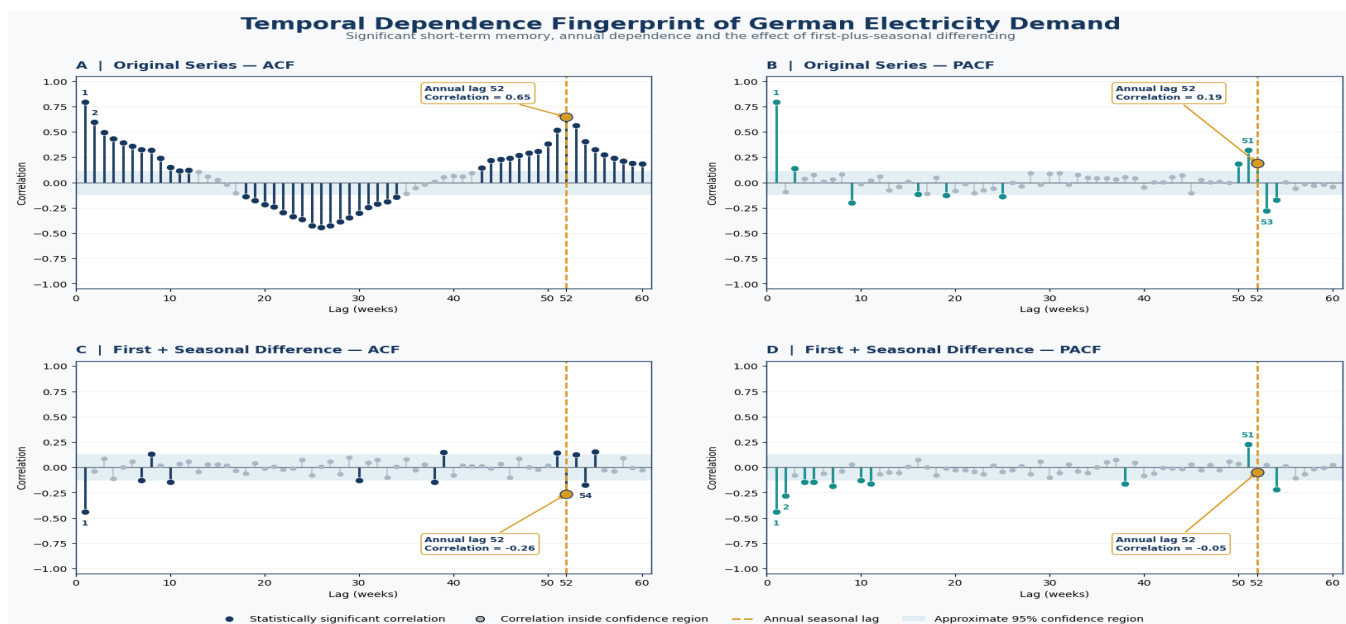


Figure 2. Correlation fingerprint before and after combined differencing. Lag 52 exposes annual dependence; the lower panels show the dependence removed by transformation.

3 FIVE COMPETING EXPLANATIONS OF DEMAND

The models were used to investigate different theories for the consumption of German power. Each model answered a particular question identified in the exploratory analysis.

Seasonal naïve Hypothesis:

- ❖ Last year's demand in the same week is still a good predictor of today's demand.
- ❖ Evidence for selection: The time series plot exhibited a repeated pattern annually and the ACF indicated a substantial dependency at lag 52.
- ❖ Reason for inclusion: It provided a robust, transparent annual standard to evaluate all advanced models.
- ❖ Main weakness: It is sluggish to adjust to structural changes, assuming the demand of the previous year stays relevant.

SARIMA Hypothesis:

- ❖ Weekly demand is predictable from linear temporal dependence and annual stochastic seasonality.
- ❖ Selection evidence: AIC grid search of 147 combinations of non-seasonal factors, followed by residual diagnostics.
- ❖ Reason for inclusion: It explored if the autocorrelation and yearly dependence found using the ACF could be modelled by a formal statistical model.
- ❖ Main weakness: The chosen model was perhaps over-differenced and had unstable coefficients, which lowered its long-horizon forecast.

SARIMAX Hypothesis:

- ❖ Temperature stress and public holidays explain extra variance in demand that demand history cannot explain by itself.
- ❖ Selection evidence: Three groups of external variables were examined using a 52-week validation period, and the best validation result was from the combined weather and holiday group.
- ❖ Reason for inclusion the preliminary investigation showed that electricity demand varies with seasonal temperature conditions and working day activity.
- ❖ Major weakness: The outcome relies on the availability of an external weather forecast, since the future temperature is not known at the forecast origin.

Random Forest Hypothesis:

- ❖ By using nonlinear interactions between demand lags, rolling statistics, calendar variables, temperature and holidays, forecast accuracy can be improved
- ❖ Selection evidence: Recursive validation was used to evaluate six hyperparameter configurations, and the final model was selected.
- ❖ Reason for inclusion: It permitted the analysis to test for nonlinear correlations without imposing a fixed linear model structure.
- ❖ Main vulnerability: In recursive forecasting, missing observations are replaced by earlier forecasts. Thus, forecast inaccuracies can be propagated into later lag and rolling features, and accumulate over time.

Recursive LSTM Hypothesis:

- ❖ Patterns acquired from hourly demand sequences can be applied to an accurate long term weekly forecast.
- ❖ Selection evidence: Comparison of three architectures utilizing validation performance and early halting.
- ❖ Reason for inclusion: The exploratory study on an hourly basis demonstrated strong intraday and weekly sequence patterns that might potentially be learned by a recurrent neural network.
- ❖ Main vulnerability: The model generated 17,472 recursive hourly forecasts. Small errors were fed back into the input sequence frequently exposing the system to bias and significant long horizon instability.

LEAKAGE CONTROLS

The first 195 weeks are the training sample, and the last 104 the unmodified test. All scalers are fit on training partitions exclusively. Features for random forests are built from demand precisely preceding each objective. Rolling windows exclude the target week. In the final forecast, a projected load is added to history before the next row is built. So realized test demand never gets into later lags. The hourly LSTM is similar: each forecast refreshes its input window of 168 hours. The drawback is more openly exposed in that the test employs realized weather, making SARIMAX a conditional forecast, rather than lag temperature regressors from the aim.

Random Forest Hyperparameter Tournament

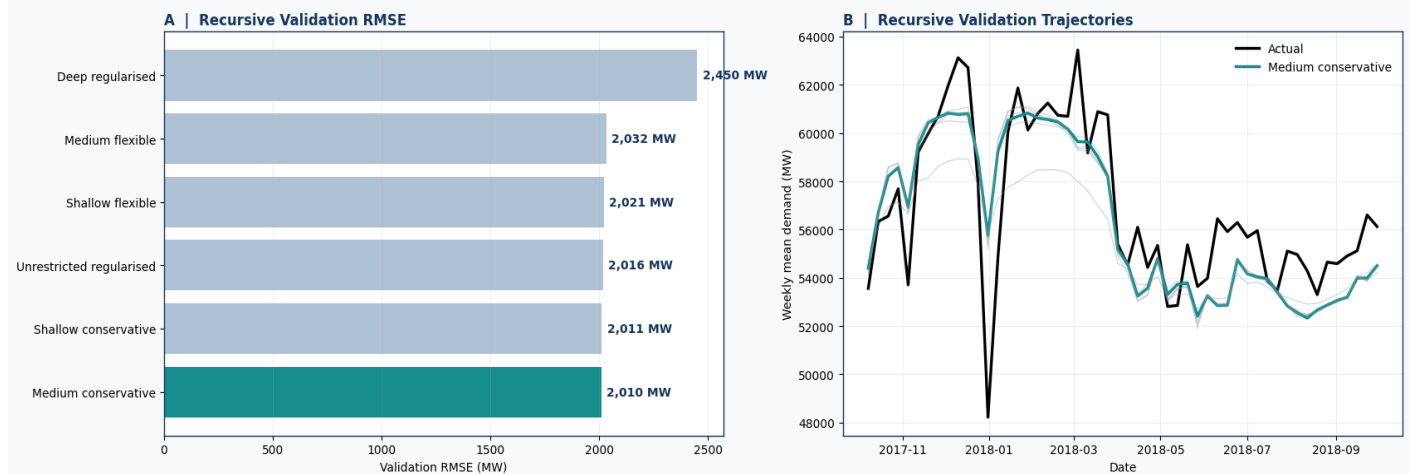


Figure 3. Random Forest hyperparameter tournament. The narrow cluster among five configurations shows that the validation conclusion is not driven by one lucky setting.

The medium conservative forest (400 trees, depth 8, minimum leaf 6 and 70% feature sampling) produced the best recursive validation RMSE, 2,010.22 MW. The LSTM search settled on a stacked 64-32-unit design with 0.20 dropout and 20 epochs but its four-week recursive validation RMSE was already 12,196.71 MW: a warning that sequence capacity did not equal steady free-running behavior.

4 THE VERDICT

Model	RMSE	MAE	MAPE	MASE	Bias	Skill
Seasonal naïve	2,988.25	2,288.49	4.36%	1.694	-1,747.80	0.00%
Random Forest	3,153.71	2,405.64	4.62%	1.781	-1,965.50	-5.54%
Mean	4,402.06	3,789.85	6.98%	2.805	-561.19	-47.31%
Drift	4,466.49	3,850.78	7.14%	2.850	-966.39	-49.47%
Naïve	4,475.77	3,858.15	7.15%	2.856	-984.53	-49.78%
SARIMAX	7,013.83	5,879.84	11.11%	4.352	-5,877.54	-134.71%
SARIMA	8,901.24	7,636.99	14.36%	5.653	-7,636.99	-197.87%
Recursive LSTM	10,236.18	9,179.25	16.15%	6.794	+9,179.25	-242.55%

Table 1. Blind 104-week performance. Errors and bias are MW; bias = actual - forecast; positive skill would indicate improvement over seasonal naïve.

The ranking is stable across RMSE, MAE and MAPE: seasonal naïve is first and Random Forest second. Because Random Forest is 5.54% worse in RMSE, its flexibility does not mean a significant improvement. The negative seasonal-naïve bias implies overprediction under the actual-minus-forecast convention, consistent with demand

dropping in the test period. The MASE for all the methods is more than one, indicating that none of the methods has a clear advantage over seasonal persistence under the scaling rule of the notebook.

Answer to Analysis Question 1 None of the models offer any notable improvement over the seasonal naïve. The closest learned option is Random Forest; however the negative skill means that the added complexity yields no net accuracy over the selected horizon.

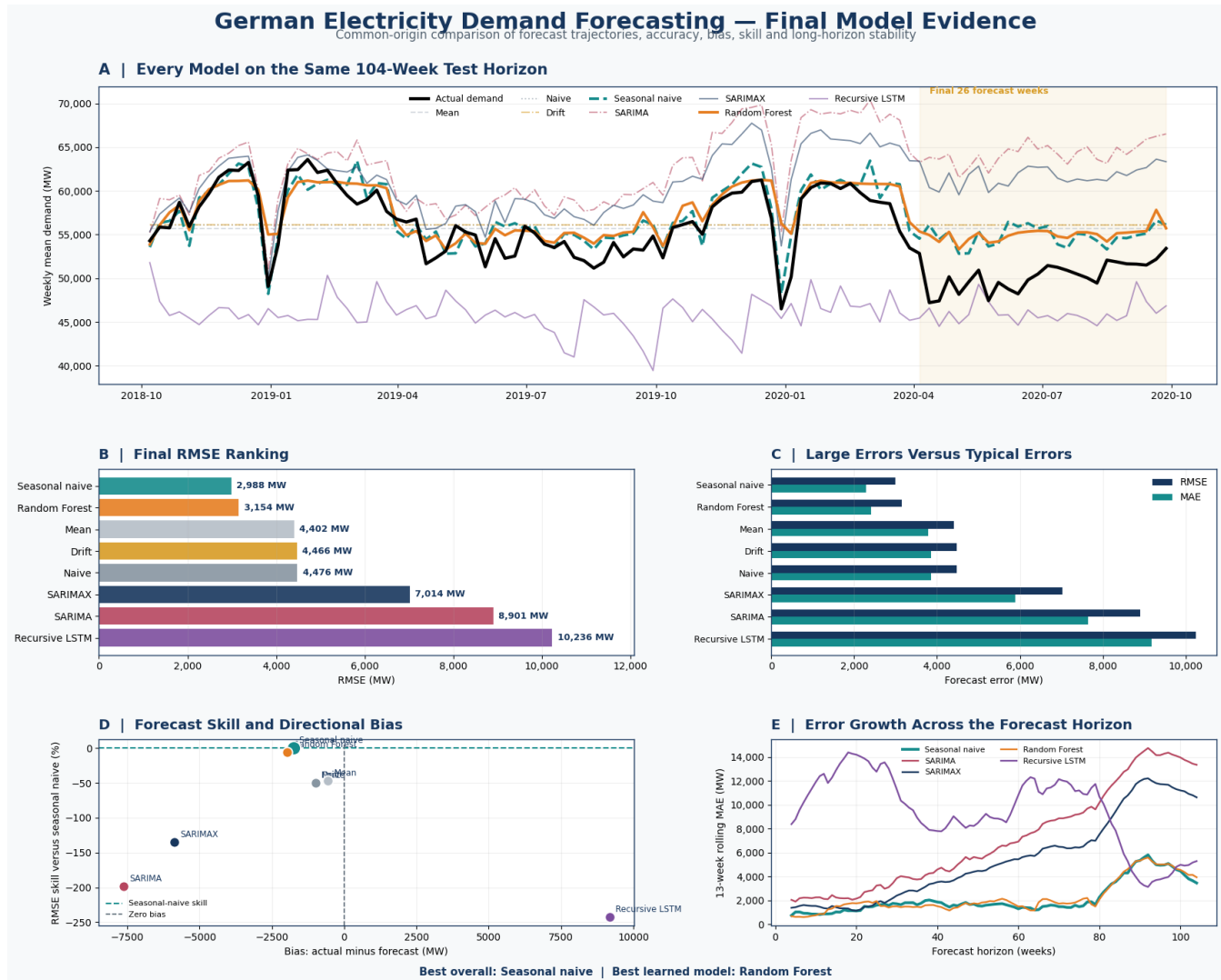


Figure 4. Final common-origin evidence dashboard: trajectories, rankings, bias, skill and error growth.

Figure 4 shows that the result is not decided by one unusually bad week. Seasonal naïve and Random Forest remain close through much of the first year, while the mean, drift and ordinary naïve forecasts occupy a higher-error middle group. SARIMA, SARIMAX and the recursive LSTM separate much earlier and their rolling errors continue to widen. The benchmark victory is therefore persistent across the horizon rather than an artefact of a single extreme event. The difference between seasonal naïve and Random Forest is small in absolute terms—165.46 MW in RMSE and 117.15 MW in MAE—but it is operationally important because the forest requires feature generation, tuning, recursive state management and retraining. That extra burden would be justified only by a repeatable gain. Instead, its -5.54% skill indicates that added flexibility slightly damages generalization. The sensible interpretation is not that Random Forest is ineffective; it is a credible challenger whose incremental value has not yet exceeded its maintenance cost. The bias-skill panel separates two different failure modes. Seasonal naïve, Random Forest, SARIMA and SARIMAX have negative bias, meaning that they systematically forecast above the realized demand level. Their seasonal shapes remain recognizable, but they adapt too slowly to the downward

level shift in the test period. The LSTM fails in the opposite direction: its +9,179.25 MW bias reflects severe underprediction as recursive feedback pulls the sequence toward an unrealistically low attractor. Consequently, averaging all models would be unsafe; opposing biases could appear to cancel in aggregate while hiding structurally unstable components. Practical verdict Seasonal naïve wins because it combines the lowest error with stable horizon behavior and minimal operational burden.

5 FORECASTS

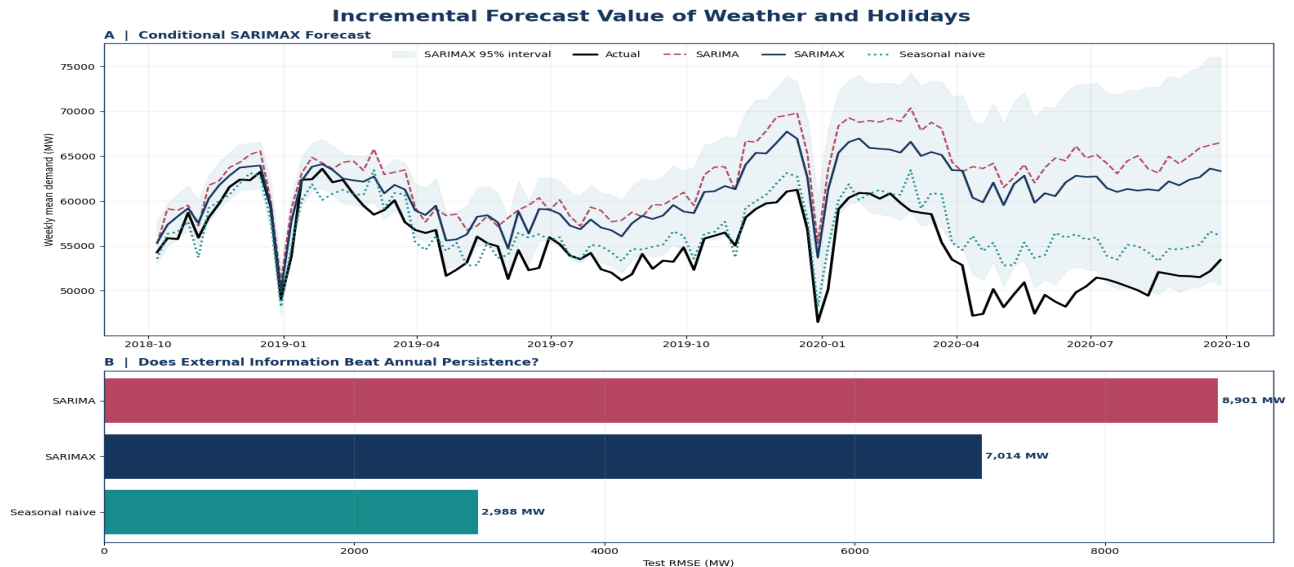


Figure 5. SARIMAX vs SARIMA and the annual benchmark

The realised test data fall below several model trajectories, particularly late in the horizon. The seasonal benchmark retains the observed annual amplitude but overpredicts level, producing bias of $-1,747.80$ MW. Random Forest also overpredicts ($-1,965.50$ MW) because recursive lag replacement carries its own higher forecasts into subsequent rolling features. Nevertheless, it remains close to the seasonal benchmark because lag 52 dominates feature importance. SARIMAX reduces RMSE from 8,901.24 to 7,013.83 MW—a 21.20% improvement over SARIMA—so temperature and holiday variables contain incremental information. The combined group was selected on validation (RMSE 2,421.11 MW). Yet both statistical models remain far behind seasonal naïve. The SARIMA search imposed $D=1$ and selected $d=2$ despite stationarity evidence; compounded differencing and a near-singular covariance matrix make its long-range level fragile. SARIMAX partly corrects the shape through degree days but cannot repair that foundation.

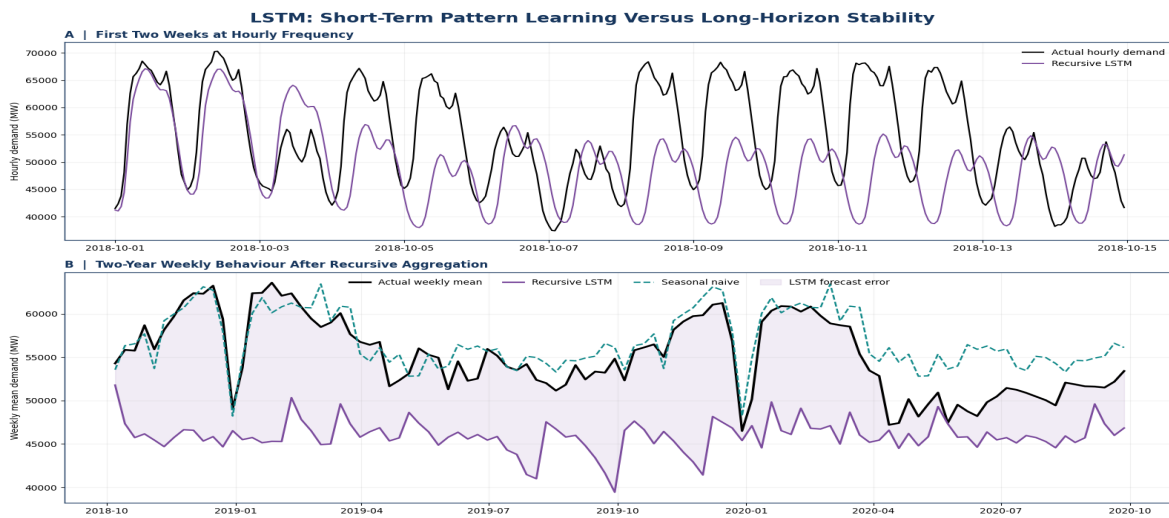


Figure 6. LSTM behaviour at hourly and weekly scales. Short-term oscillation is learned, but the recursive two-year path settles at an incorrect level.

The first hours of the LSTM are like the observed intraday cycle, suggesting that the network learnt local sequence structure. That success doesn't survive 17,472 recursive steps. The weekly RMSE is 10,236.18 MW and the positive bias of 9,179.25 MW shows a substantial underprediction. The failure mechanism is exposure bias: the inputs are seen at training, but at deployment the inputs are more and more the model's own faulty outputs. SARIMA and SARIMAX intervals are useful for diagnostics, but not decision-grade yet. Parameter instability, structural change and conditional weather inputs can cause nominal intervals to be overconfident. Operational reserve planning should consequently be grounded in empirically recalibrated coverage, rather than in notional confidence labels

6 OUTCOMES & FUTURE WORK

1. The analysis is based on 50,400 hourly observations of German power demand from January 2015 to September 2020. The timestamps were checked, duplicates removed, invalid demand values screened, and incomplete weeks excluded, resulting in 299 complete weekly mean-demand observations averaged hourly. The first 195 weeks were used for model development while the latter 104 weeks were held out as a blind two year test phase. Exploratory investigation showed a strong annual cycle between winter and summer, decreased weekend consumption, a clear intra-day profile and a reduction in the level of demand throughout the later test period. These properties encouraged the comparison of seasonal benchmarks, statistical models, a feature-based Random Forest and an hourly LSTM.
2. A significant finding was obtained from the stationarity testing. The weekly series was stationary as shown by both ADF ($p = 0.0020$) and KPSS ($p = 0.1000$). Stationarity was achieved by first-differencing and by combination first- plus seasonal differencing, while seasonal differencing alone failed both tests. But the ACF exhibited a substantial dependency on or around lag 52 which was unambiguous evidence of annual persistence. The observed demand structure was consistent with a seasonal length of 52 weeks. But stationarity evidence also indicated that considerable differencing was not required and could result in unstable long-range forecasts.
3. The seasonal naïve model scored the best overall test accuracy with RMSE of 2,988.25 MW, MAE of 2,288.49 MW and MAPE of 4.36%. The Random Forest was the most powerful machine-learning alternative, with RMSE 3,153.71 MW and MAPE 4.62%. Its RMSE was just 165.46 MW greater than seasonal naïve, but its forecast skill was -5.54%, which showed that the model did not improve upon the benchmark in a meaningful way. The mean, drift and ordinary naïve forecasts were in a middle-performing group because they were not able to replicate annual demand fluctuation as well as the seasonal benchmark.
4. The SARIMA grid search adopted a seasonal specification of $(1, 1, 1)[52]$ and examined 147 candidate ordinary orders. AIC selected SARIMA $(2, 2, 6) \times (1, 1, 1)[52]$, yet the test RMSE of the model was 8901.24 MW and the forecast skill was -197.87%. Its negative bias of -7,636.99 MW was a serious overprediction. The poor outcome was consistent with likely over-differencing, a near-singular covariance matrix and lower demand during the test time. This showed that the model chosen by the lowest training AIC is not necessarily the best performing when forecasting on new data.
5. SARIMAX did better than SARIMA in adding temperature and holiday variables. The weather-and-holiday feature group together obtained a validation RMSE of 2,421.11 MW, while the final SARIMAX RMSE was 7,013.83 MW. This was a 21.20% improvement over SARIMA but still much worse than seasonal naïve. Holidays are known at the prediction origin but temperature across the full two-year horizon is not. The SARIMAX result should be considered as a conditional situation because the test-period temperature was used as a realised value. A deployable version would need historical weather forecasts or a different weather forecast model.

6. The best Random Forest had 400 trees, a maximum depth of eight, a minimum leaf size of six and 70% feature sampling. Its recursive validation RMSE was 2010.22 MW. Lagged demand, rolling summaries, yearly memory, calendar impacts, temperature and holiday variables allowed the model to mimic much of the seasonal pattern. But it failed to surpass seasonal naïve as recursive prediction replaced observed lag values with earlier forecasts step-by-step. Thus, errors were introduced in later feature rows and accumulated along the horizon. However, the extra feature engineering, calibration and maintenance of the model is not warranted without repeated positive skill.
7. The chosen LSTM has stacked layers of 64 and 32 units with 0.20 dropout and was trained for 20 epochs. It learned the short-term hourly demand cycle but failed in the 17,472-step recursive forecast. Its last weekly RMSE was 10,236.18 MW, MAPE was 16.15% and forecast skill was -242.55%. Positive bias of 9,179.25 MW indicated serious underprediction. The key difficulty was exposure bias. The network was trained with observed sequences but was expected to forecast using more fake inputs. Consequently, local hourly pattern detection did not convert into solid long-term predictions.
8. All leak-sensitive operations were ordered chronologically. The test time was not included in training, scaling, feature selection and hyper parameter tuning. Lagged demand and rolling statistics were built using only observations prior to the target. In recursive forecasting, the model projections were used to substitute the unavailable future demand rather than the realized test values. Demand leakage was prevented but the SARIMAX evaluation is still dependent on realized future weather. The temperature constraint was explicitly disclosed:

Thus, the operational approach is to use seasonal naïve as the primary two-year, weekly point forecast. It is the most accurate, has the most consistent horizon behavior and the lowest maintenance burden. Its negative bias should be watched because it responded slowly to the lower demand level during the test period. Random Forest should stay a challenger model and SARIMAX can accommodate conditional weather scenarios. Current SARIMA and recursive LSTM standards need a major modification before use. Future work should consider rolling-origin testing for several forecast horizons instead of the single-origin evaluation. SARIMA should be revisited without forced regular differencing. ETS and Fourier dynamic-regression models should be tested. Weather analysis should be based on archival forecasts and population-weighted regional observations, rather than realized Berlin temperature alone

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